SUSTAINABLE AGRICULTURE SUMMIT * CASE STUDY BRIEFING

ECOLAKESIDE FARM INITIATIVE

Master Environmental Farm Plan: 240-Hectare Sustainable Companion Farming & Watershed Defense

Potatoes & Garlic Companions | **4R Nutrient Stewardship** | **Constructed Wetland Bio-Filter** | **Nutrients For Life Partner**

240 ha	**210 ha**	**30 ha**	**0 Litres**
Total Landscape Area	**Companion Cropland**	**Water Buffer & Zones**	**Lake Extraction**
Whole Property	Potatoes & Garlic	Wetlands & Forest	100% Recycled Drip

1. HEADER & NAVIGATION STRUCTURE

- Brand Title: EcoLakeside Farm
- **Navigation Sections:**

1. Intro (`#hero`)
2. Limitations (`#overview`)
3. Plan (`#masterplan`)
4. Environment (`#soil-water`)
5. Off-Season (`#alternate-uses`)
6. Recap (`#recap`)
7. Sources (`#citations`)

- Presentation Controls: 6-Part Guided Slide Mode with real-time visual progress slider.

2. PART 1: INTRODUCTION (HERO SECTION)

Eyebrow Label

Sustainable Agriculture Summit * Case Study Presentation

Main Title

EcoLakeside Farm Initiative

Subtitle

Balancing high-yield crop production, watershed health, and long-term economic resilience on a 240-hectare landscape.

Landscape Summary Metrics

- 240 ha: Total Landscape Area
- 210 ha: Cultivated Companion Cropland (Potatoes & Garlic)
- 30 ha: Dedicated Ecological Water Buffer & Filtration Zones

- 0 L: Zero Direct Water Extraction from Freshwater Lake

3. PART 2: LIMITATIONS & SITE CHALLENGES

The Planning Mandate

Our 240-hectare farmland has rich potential, but also four major physical limitations that demand smart agricultural design:

1. Sandy Loam Soil (pH 6.0-7.0): Sandy soil is loose and easy for root expansion, but drains water rapidly and holds few nutrients. Without precautions, dry spells cause drought stress, while heavy rains can wash nitrogen straight down toward groundwater.
2. Eastern Downward Slope: The land slopes directly toward a freshwater lake on the east. Rain easily picks up speed downhill, carrying loose topsoil and fertilizer directly toward the lake.
3. Drought & Moisture Vulnerability: Because water perks through sandy loam so fast, summer dry spells can quickly deplete crop root moisture without active retention strategies.
4. Sensitive Freshwater Lake: Excess nitrogen and phosphorus runoff triggers toxic blue-green algae blooms, depleting oxygen and harming fish. Our plan guarantees zero unfiltered runoff reaches the lake.

Our Integrated Solution

By pairing Potatoes with Garlic in a companion planting system, farming perpendicular to the slope (across contours) to stop runoff, and routing drainage through a 3-stage biological water-cleaning system, we protect the lake while maximizing farm profitability.

Site Constraints & Built-In Defenses Matrix

- **Perpendicular Contour Farming:**

Tillage rows run across the slope (perpendicular to water flow). Each furrow acts as a mini-dam that traps rainwater in place, stops downhill erosion, and protects valuable topsoil.

- **Drought Precautions & Soil Retention:**

Adding loamy biochar and composted manure boosts the soil's sponge capacity. Drip irrigation from our recycling ponds ensures crops never thirst during dry prairie weeks.

- **Lake Runoff & Water Cleaning:**

A 30-meter tree buffer and 3-stage wetland system trap sediment and naturally filter out 88%+ of dissolved nutrients before clean water ever returns to the water supply.

- **Potatoes + Garlic Synergy:**

Garlic naturally deters Colorado potato beetles and soil pests with its sulfur aroma, eliminating harsh chemical sprays while producing two high-value food crops.

4. PART 3: THE 240-HECTARE FARM MASTER PLAN

Spatial Zoning & GIS Layout

- 01. Potato Production Zone (100 Hectares / 41.7% Area) *Topography:* Upper Slope / Western Elevation *Soil Type:* Sandy Loam, Target pH 6.0 to 6.5 *Rationale:* Located on the highest ground of the property to eliminate the hazard of waterlogging and root rot. Sandy loam provides loose, friable pore space required for unrestricted tuber expansion, uniform sizing, and clean mechanical harvesting without soil clods. *Lake Protection:* Contour tillage furrows run perpendicular to the eastern slope, trapping rainfall in micro-reservoirs. A winter rye cover crop is seeded immediately following early harvest to anchor sandy loam against winter wind and spring melt erosion. *Pest Protection:* Garlic companion rows emit pungent sulfur volatiles that naturally mask potato scent and repel Colorado potato beetles. *Pollinator Allies:* Native bumblebees (*Bombus*) provide sonic "buzz pollination" that vibrates blooms and strengthens fruit/seed sets. *Drought Precautions:* Loamy biochar and composted manure boost moisture holding, while precision drip irrigation delivers recycled pond water.
- 02. Hardneck Garlic Companion Zone (110 Hectares / 45.8% Area) *Topography:* Mid to Lower Slope Companion Plain *Soil Type:* Sandy Loam with Organic Matter, pH 6.2 to 6.8 *Rationale:* Planting garlic alongside potatoes provides world-class biological companion benefits. Garlic releases pungent sulfur volatiles that naturally mask potato scent and repel Colorado potato beetles, aphids, and flea beetles, eliminating synthetic insecticide sprays. *Lake Protection:* Garlic's moderate nutrient requirements and compact root network reduce nitrogen leaching. Planting across the slope perpendicular to runoff creates living barriers that hold water and stop topsoil loss. *Natural Insect Shield:* Repels Colorado potato beetles and soil pests naturally without chemical pesticides (supported by Gilmour Companion Planting Guide). *Staggered Harvest:* Harvested in July/August, allowing early summer cover crops (clover, buckwheat) to restore organic matter before winter. *High Market Value:* Generates $8-$14/kg farmgate revenue, providing premium economic stability alongside potato staple production.
- 03. Ecological Water Filtration System (~5 Hectares / 2.1% Area) *Topography:* Northeast Topographic Catchment *Soil Type:* Terraced Constructed Wetland and Settling Ponds *Rationale:* A three-stage biological filtration complex situated at the natural drainage convergence in the northeast corner. It captures agricultural surface runoff and tile drainage before it reaches the lake. *Lake Protection:* Stage 1 settles suspended sands and silts. Stage 2 utilizes native bulrushes and cattails for vegetative phosphorus and nitrogen bio-absorption. Stage 3 stores polished, stabilized water for irrigation recycling. *Performance:* Binds particulate and dissolved reactive phosphorus, removing triggers for blue-green algae, and recovers snowmelt for closed-loop crop irrigation.
- 04. 30-Meter Natural Riparian Buffer (~20 Hectares / 8.3% Area) *Topography:* Eastern Lake Shoreline Margin *Soil Type:* Native Riparian Alluvial Forest Profile *Rationale:* A continuous 30-meter wide non-cultivated multi-canopy buffer extending along the full eastern shoreline. Designed according to provincial riparian health standards. *Lake Protection:* Combines deep-rooted willows, trembling aspen, red-osier dogwood, and native prairie grasses. This living vegetative barrier traps over 85% of particulate sediment and stabilizes the shore against wave erosion. *Wildlife Corridor:* Maintains an unbroken migratory corridor for waterfowl, bumblebee pollinators, and predatory insects.
- 05. Protected Wetland Reserve (~5 Hectares / 2.1% Area) *Topography:* Southeast Shoreline Sanctuary *Soil Type:* Hydric Wetland Peat and Saturated Silt *Rationale:* An undisturbed natural wetland retained in its native state, functioning as a permanent biodiversity refuge and hydrological surge buffer. *Lake Protection:* Absorbs seasonal storm surges and regulates groundwater recharge into the lake without thermal or chemical disturbance. Maintained under a strict zero-chemical and zero-machinery exclusion perimeter.
- 06. Perpendicular Contour Farming (Entire Cultivated Landscape / 210 Hectares) *Topography:* Full Eastern Descending Slope *Soil Type:* Contoured Sandy Loam Micro-Terraces *Rationale:* All field operations, planting rows, and cultivating passes are oriented along precise contour lines perpendicular to the slope descending toward the lake. *Lake Protection:* Every raised furrow acts as a natural mini-dam. This cuts surface water runoff velocity by over 70%, trapping water in place, preventing soil rills, and protecting valuable topsoil.

5. PART 4: SOIL DYNAMICS, WATER CLEANING & ENVIRONMENTAL DEFENSE

Pillar A: Sandy Loam Care & Drought Defense

- **Perpendicular Farming Across the Slope:**

Rather than tilling downhill toward the lake, our tractor passes run strictly across the slope (perpendicular to water flow). Each raised furrow acts like a miniature dam, catching rainfall where it falls, stopping fast runoff, and keeping fertile topsoil on the field.

- **Drought Precautions & Biochar Sponge:**

Coarse sandy soil dries out quickly during prairie summers. We add loamy biochar and composted organic matter to dramatically boost the soil's water-holding capacity, keeping root zones moist for days longer during dry spells.

- **Sustainable Manure Used Away from Water:**

We recycle composted manure from regional livestock farms to feed soil biology and build humus naturally. Crucially, manure is applied only to upper inland zones---strictly 150+ meters away from the riparian lake buffer---so no animal waste or nutrients ever touch the water.

- **Winter Cover Crops to Lock Soil:**

Right after our potato and garlic harvests, we plant cold-hardy winter rye. The rye's thick roots bind loose sand throughout freezing winters and spring thaws, stopping wind and meltwater erosion completely.

Pillar B: Water Quality, Cleaning & Biological Allies

- **Cleaning the Water Supply (3-Stage Filtration):**

Field runoff doesn't reach the lake. Instead, it enters a natural cleaning circuit: settling out silt in a sediment trap, filtering through wetland cattails and bulrushes that absorb dissolved nitrogen and phosphorus, and storing crystal-clean polished water in our retention pond.

- **Closed-Loop Irrigation (Zero Direct Lake Draw):**

Our precision drip and pivot irrigation systems draw 100% of their water from our recycled retention ponds. We take zero water from the natural lake, protecting fish spawning beds and shoreline water levels.

- **Bumblebee Presence & Pollination Benefits:**

Our 30-meter tree buffer and wildflower strips create permanent homes for native wild bumblebees (Bombus). Bumblebees provide specialized "buzz pollination" (vibrating blossoms with wing muscles), improving potato flowering vigor and keeping the local ecosystem healthy and resilient.

- **Natural Predators Stop Colorado Potato Beetles:**

Instead of spraying harmful insecticides, we let nature work for us. Wildflower borders shelter ladybugs, green lacewings, predatory stink bugs, and ground beetles that eagerly eat Colorado potato beetle eggs and larvae, keeping crops healthy without toxic chemicals.

Stopping Leaching & Groundwater Contamination

- **Subsurface Banding:**

Fertilizer is placed in tight bands 2 inches beside and below seeds, right where roots can drink it. No surface broadcasting that could wash away in rainfall or blow away in the wind.

- **Controlled-Release Nutrients:**

We use polymer-coated granules that slowly feed nitrogen as soil warms, matching crop growth speed so excess fertilizer never sits loose in sandy soil.

- **Groundwater Protection:**

Split feeding during peak tuber growth and deep-rooted cover crops prevent nitrate leaching, keeping the underground drinking water table 100% pure and safe.

The 3-Stage Ecological Water Filtration Process

1. Stage 01 --- Sedimentation Basin: Gravity naturally settles out sand, silt, and loose soil particles before water moves forward. Result: 75%+ of suspended soil trapped.
2. Stage 02 --- Wetland Bio-Filter: Native cattails, rushes, and root microbes naturally consume and filter out dissolved nitrates and phosphorus. Result: 88%+ of dissolved nutrients absorbed.

3. Stage 03 --- Polishing & Drip Reuse: Purified, clean water rests in our retention basin, ready to be pumped back through high-efficiency crop drip lines. Result: 100% recirculated closed-loop water; zero lake extraction.

6. CROP SELECTION RATIONALE: POTATOES & GARLIC COMPANION CROPPING

Crop 1: Russet & Table Potatoes (100 ha / 41.7%)

- Topographic Placement: Upper Western Slope ensures gravity drainage, preventing waterlogging, tuber rot, and late blight.
- Soil Compatibility: Loose sandy loam allows uniform, unhindered tuber expansion without physical soil clods. Maintaining pH below 6.5 naturally suppresses potato scab (*Streptomyces scabies*).
- High Food Energy Density: Yields up to 15 million calories per hectare, loaded with potassium, vitamin C, dietary fiber, and healthy carbohydrates.

Crop 2: Hardneck Garlic Companion (110 ha / 45.8%)

- Natural Insect Shield Against Colorado Beetles: Pungent sulfur aroma masks potato plants and naturally deters Colorado potato beetles, flea beetles, and aphids without synthetic chemical pesticides (supported by Gilmour Companion Guide).
- Non-Competing Root Zones: Compact, shallow root networks do not compete with potato tubers for root space or deep moisture. Releases natural anti-fungal sulfur compounds into surrounding soil.
- Staggered Harvest Timing: Harvested in late July to early August (weeks before potatoes), enabling immediate summer cover cropping with buckwheat or clover to rebuild soil carbon.
- High Economic Value: Farmgate garlic commands $8-$14/kg, providing high-margin cash flow.

Three-Year Companion Crop Rotation Cycle

- Year 1: Potatoes + Winter Rye Cover Crop: High-yield potato harvest followed immediately by drilled winter rye to anchor topsoil through freezing prairie winters.
- Year 2: Garlic + Summer Green Manure: Garlic planted in autumn and harvested mid-summer, followed immediately by nitrogen-fixing clover and buckwheat to restore soil health.
- Year 3: Legume / Brassica Bio-Fumigant: Peas, alfalfa, or mustard crop fixes atmospheric nitrogen, breaks all potato pathogen cycles, and revitalizes beneficial soil microbes.

7. FOOD & WATER SECURITY FRAMEWORK

Food Security Architecture

- 15 Million Calories per Hectare: High caloric density per hectare outyielding cereal grains in usable dietary energy.
- High-Value Specialty Garlic Market: Diversifies revenue, protects against potato commodity swings, and offers 6 to 9 months of shelf-stable storage without energy-intensive refrigeration.
- Long Shelf-Life Food Reserves: Provides dependable regional food security for Manitoba and western Canada throughout winter and spring.

Water Security Architecture

- 180,000 m^3 Stormwater & Snowmelt Harvesting: Captured in retention ponds to guarantee ample irrigation throughout dry prairie summers.
- Zero Direct Lake Extraction: Protects littoral freshwater fish habitats and natural shoreline water levels.
- 95% Efficient Subsurface Drip Irrigation: Delivers recycled water directly beneath crop roots, eliminating the 30%+ evaporative loss of traditional overhead sprayers.

8. PART 5: OFF-SEASON AGRITOURISM, WINTER PARK & MONETIZATION

Autumn (Sept-Oct): Harvest Market & Farm Days

- Direct farmgate festival celebrating fresh potato and garlic harvests.
- Heirloom pumpkin patches, garlic braiding workshops, and interactive school tours explaining 4R soil stewardship and living soil biology.
- Immediate post-harvest cover crop seeding across all fields.

Winter (Nov-Jan): Holiday Light Drive & Frozen Lake Skate Park

- Drive-Through Light Route: A 4 km illuminated holiday light festival placed **strictly along the central road dividing the potato and garlic fields, continuing in a continuous scenic loop all around the farm's outer perimeter roads**.
- Frozen Lake Skate Park: Designated groomed ice skating park on the safely frozen lake near the shoreline, complete with festive music, light arches, and heated equipment sheds.

Winter Park Monetization Strategy (~$215,000 Net Projected Revenue)

1. $25 / Vehicle Admission: Drive-through light show ticket (~5,000 vehicles = $125,000).
2. $10 / Person Lake Skate Pass: Daily wristband for the illuminated lake ice park (~6,000 skaters = $60,000).
3. $8 / Pair Skate & Helmet Rentals: On-site equipment rentals in warm barn outfitter stations (~2,500 rentals = $20,000).
4. Vendor Stall Leases & Concessions: Heated barn stalls leased to local bakeries, hot chocolate/cider stands, plus evening fire pit rentals ($15,000+).

- Economic Impact: Transforms the dormant winter season into positive cash flow, provides local seasonal employment, and supports regional small businesses while fields rest.

Long-Term De-Commissioning Contingency Plan

If economic conditions or environmental mandates ever necessitate retiring the land from commercial crop production, it transitions into an Agrivoltaic Solar Eco-Reserve:

- 45 MW Clean Renewable Electricity: Elevated bifacial solar panels allow sheep grazing and pollinator wildflower meadows beneath.
- Permanent Conservation Easement: Enrolled with ALUS and the Nature Conservancy of Canada, permanently safeguarding the lake watershed from any industrial or urban disturbance.

9. 4R NUTRIENT STEWARDSHIP CERTIFICATION (NUTRIENTS FOR LIFE CANADA)

EcoLakeside Farm adheres strictly to Fertilizer Canada's science-based 4R Nutrient Stewardship framework, supported by Nutrients for Life Canada:

- **Pillar 01 --- Right Source:**

Polymer-coated controlled-release urea (ESN) releases nitrogen gradually as soil warms. Composted manure applied strictly to upper inland fields feeds soil biology sustainably without runoff risk.

- **Pillar 02 --- Right Rate:**

Precision GPS soil sampling calibrated specifically to potato tuber demand and garlic sulfur requirements. Variable-Rate Application prevents over-fertilizing sandy ridges.

- **Pillar 03 --- Right Time:**

Split feeding: starter at planting and fertigation during active tuber bulbing. Strict zero application on frozen, snow-covered, or waterlogged ground.

- **Pillar 04 --- Right Place:**

Subsurface precision banding 2 inches below and beside seeds. Zero surface broadcasting near the 30-meter riparian lake buffer or water channels.

10. PART 6: EXECUTIVE RECAP & KEY TAKEAWAYS

1. High-Yield Companion Crops

Potatoes (100 ha) provide 15M cal/ha of staple food, while companion Garlic (110 ha) naturally repels Colorado potato beetles and soil fungi with sulfur vapors.

Key Metric: Zero reliance on chemical insecticides.

2. Closed-Loop Lake Protection

Perpendicular contour farming stops downhill erosion. A 3-stage constructed wetland cleans 100% of field runoff for drip irrigation. Zero direct lake pumping or pollution.

Key Metric: 88%+ nutrient absorption & clean recycled water supply.

3. Year-Round Monetization

Off-season drive-through holiday light festival between crop rows and along perimeter roads, plus a frozen lake skate park, generating over $200k in profitable winter revenue.

Key Metric: Continuous seasonal cash flow and community engagement.

Summit Takeaway

> *"Agriculture that enriches the land rather than depleting it." By harnessing natural biology---bumblebee pollination, predatory insects, biochar moisture buffers, composted manure, and 4R precision banding---EcoLakeside Farm proves high yields and watershed protection thrive together.*

11. REFERENCES & DIRECT SOURCE LINKS

1. Delta Farmland: Winter cover cropping and farm soil protection guide [https://deltafarmland.ca/cover-crop-for-winter-how-to-protect-farm-soil/](https://deltafarmland.ca/cover-crop-for-winter-how-to-protect-far
2. Gilmour: Companion planting chart and agronomic compatibility guide https://gilmour.com/companion-planting-chart-guide/
3. Fertilizer Canada: 4R Nutrient Stewardship across Canadian agriculture [https://fertilizercanada.ca/our-focus/stewardship/4rs-across-canada/](https://fertilizercanada.ca/our-focus/stewardship/4rs-across-canad
4. Water Security Agency (WSask): Prairie watershed management and agricultural water stewardship https://wsask.ca
5. Wakefield Biochar: Loamy soil conditioning, moisture retention, and microbial ecology [https://wakefieldbiochar.com/learning-center/loamy-soil-conditions/](https://wakefieldbiochar.com/learning-center/loamy-soil-conditions/

12. FOOTER & PARTNER RECOGNITION

- Project: EcoLakeside Farm Initiative (Team 4 Case Study)
- Summit: "Sustainable Agriculture - Challenges, Opportunities, and Implications" Summit
- Educational Partner: Nutrients For Life Canada (https://nutrientsforlife.ca/en)

- Presentation Duration: Structured for a 5-minute case study pitch
- Copyright: (C) 2026 Team 4. All rights reserved.